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Pneumatic Rifle for Alternative Ammunition

Abstract

A hand-held gun for interchangeably propel a round of one of solid and liquid consistency is provided. The gun includes a gas actuator, a firing module, a magazine, a loader and a barrel. The gas actuator stores and releases compressed gas. The firing module includes a chamber for receiving the gas to push forward a bolt in response to actuation by a trigger. The module is disposed forward of the actuator. The magazine contains ammunition that constitutes the round. The loader laterally receives the round from the magazine and longitudinally receiving the bolt to eject the round. The loader is disposed forward of the module. The barrel directs the round ejected from the loader, and is disposed forward of the loader.

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Background/Summary

BACKGROUND

[0002] The invention relates generally to pneumatic or other gas-powered guns. In particular, the invention relates to a hand-held long-barrel firearm that dispenses either non-lethal solid rounds or liquid slugs depending on the selected magazine being attached for loading ammunition.

SUMMARY

[0003] Conventional gas powered rifles yield disadvantages addressed by various exemplary embodiments of the present invention. In particular, various exemplary embodiments provide a hand-held gun for interchangeably propel a round of one of solid and liquid consistency. The gun includes a gas actuator, a firing module, a magazine, a loader and a barrel. The gas actuator stores and releases compressed gas.

[0004] The firing module includes a chamber for receiving the gas to push forward a bolt in response to actuation by a trigger. The module is disposed forward of the actuator. The magazine contains ammunition that constitutes the round, whether solid round or liquid slug. The loader laterally receives the round from the magazine and longitudinally receiving the bolt to eject the round. The loader is disposed forward of the module. The barrel directs the round ejected from the loader, and is disposed forward of the loader.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] These and various other features and aspects of various exemplary embodiments will be readily understood with reference to the following detailed description taken in conjunction with the accompanying drawings, in which like or similar numbers are used throughout, and in which:

[0006] FIG. 1 is a perspective assembly view of a first embodiment of an exemplary pneumatic rifle;

[0007] FIGS. 2A and 2B are perspective assembly views of a second embodiment of an example pneumatic rifle in a first configuration;

[0008] FIG. 3 is a perspective exploded view of the second embodiment;

[0009] FIG. 4 is a perspective exploded view of the forward mid-section;

[0010] FIG. 5 is a perspective exploded view of the aft mid-section;

[0011] FIG. 6 is a perspective assembly view of the second embodiment in a second configuration;

[0012] FIGS. 7A, 7B and 7C are elevation cross-section views of the first embodiment;

[0013] FIGS. 8A and 8B are elevation cross-section views of the second embodiment in the first configuration;

[0014] FIGS. 9A, 9B and 9C are detail cross-sectional elevation views of the second embodiment in the second configuration; and

[0015] FIGS. 10A, 10B and 10C are detail cross-sectional elevation views of the second embodiment featuring the firing mechanisms.

DETAILED DESCRIPTION

[0016] In the following detailed description of exemplary embodiments of the invention, reference is made to the accompanying drawings that form a part hereof, and in which is shown by way of illustration specific exemplary embodiments in which the invention may be practiced. These embodiments are described in sufficient detail to enable those skilled in the art to practice the invention. Other embodiments may be utilized, and logical, mechanical, and other changes may be made without departing from the spirit or scope of the present invention. The following detailed description is, therefore, not to be taken in a limiting sense, and the scope of the present invention is defined only by the appended claims.

[0017] The disclosure generally employs quantity units with the following abbreviations: length in

meters (m) or inches ("), mass in grams (g) or pounds-mass (lbm), time in seconds(s), angles in degrees (°), force in newtons (N), pressures in pascals (Pa) or pounds-per-square-inch-gage (psig), temperature in kelvins (K), and energy in joules (J). Supplemental measures can be derived from these, such as density in grams-per-cubic-centimeters (g/cm.sup.3), moment of inertia in kilogram-square-meters (kg-m.sup.2) and the like.

[0018] The exemplary small Universal Dispersing System (sUDS) rifle uses gas pressure to interchangeably propel either a solid round or a finite quantity of liquid consistency, depending on the magazine attached. The sUDS rifle employs an electric switch to activate the gas release in response to pressing a trigger.

[0019] The intent of exemplary embodiments is to foster a non-lethal and/or a counter small Unmanned Aerial System (sUAS) weapon. In its present configuration, the system is designed as a rifle. However future disclosures expect modification of the single barrel rifle configuration into a multi-barrel turret.

[0020] FIG. 1 shows a perspective view **100** of a first configuration of an assembly **110** of the exemplary sUDS rifle. The assembly **110** includes an ammunition magazine **120**, a butt stock **130**, a gas actuator **140**, an upper grip cover **150**, a lower assembly **160**, an upper box receiver **170**, a lower box receiver **180** and a barrel **190**. The upper receiver **170** includes a top protective cover. The barrel **190** is presumed to include a rifled bore to provide axial spin to the projectile to be ejected. However, the disclosure conceptually includes a smooth-bore long-gun.

[0021] FIG. 2 shows a perspective view **200** of a second configuration of an assembly **210** of the exemplary sUDS rifle. The assembly **210** includes a magazine **220** that combines with the assembly **210** as a loaded sUDS rifle **230**. The actuator **140** comprises an annular compressed air container **240** and an air reservoir **245**. The container **240** can store air or an alternative compressed gas. The assembly **210** further includes the actuator **140**, an upper half-cylinder **250**, a lower half-cylinder **260**, a receiver **270**, a flange connector **280** and an air manifold **290**. The lower half-cylinder **260** includes a handle. The receiver **270** includes a protective top cover. The barrel **190** extends forward from the flange connector **280**.

[0022] FIG. 3 shows a perspective exploded view of the second configuration of components **310** for the loaded sUDS rifle **230**. The components include a bolt carrier **320**, a cradle **330**, an air loader **340** (for cycling), a serrated hollow bolt **350**, a loader **360**, solid ammunition rounds **370**, a magazine lid **380** and outer threads **390** on the barrel **190**.

[0023] The length of the rifle **110**, **230** approximates that of a Remington **870** shotgun, roughly 36 inches long, depending on the barrel **190**. The bore can be $\frac{3}{4}$ " diameter, although alternate sizes can be incorporated, corresponding with the diameter of the bolt **350**. The rifle's total weight is about ten pounds-mass.

[0024] FIG. 4 shows an exploded detail perspective view **400** of components **310** adjacent the magazine **180**, which includes a lip **410**, the loader **360** with a chamber **420**, and the flange **280** with a through-hole **430** designed to receive threads **390**. FIG. 5 shows an exploded detail perspective of aft mid-section behind the magazine **180**. This includes a manifold **510** with an end flange **515**, an annular sleeve **520**, a fore flange **530**, a cylinder plug **540**, bolts **545**, a ring **550**, a handle **560** with trigger guard **565**, a tab **570** extending from the bolt **350**, a through-hole cavity **580** of the bolt carrier **320**. The air container **240** comprises the manifold **510** and flange **515**. The air reservoir **245** comprises the sleeve **520**, flange **530** and internal components contained therein.

[0025] FIG. 6 shows a perspective view **600** of an assembly **610** the sUDS rifle **230** in the liquid firing configuration. A receiver **620** receives a detachable magazine **630** containing liquid for ammunition that can be refilled via a fluid inlet **640** as a lateral orifice. A flexible conduit (not shown) can connect to the inlet **640** to continually fill the magazine **630**. A trigger **650** within the handle **560** behind the guard **565** enables release of the metered liquid ammunition by gas ejection.

[0026] The only difference between rifle **110** and rifle **230** is the receiver **170** & **180**, **270** & **280** and the magazine **220**, **630**, depending on whether ammunition is solid or liquid. The operating

pressure, length, bore, mass do not vary between these rifles **110, 230**. Chamber pressure ranges between **100** psig and **300** psig, with muzzle exit velocity depending on that pressure. Higher chamber pressures enable increased range, but also increase the probability of lethality.

[0027] FIGS. 7A, 7B and 7C show detail cross-section elevation views of the sUDS rifle assembly **110** in the first configuration. A cylindrical annulus **710** enables a piston **720** to move aft. Aft and fore blocks **730** and **735** axially flank the cylinder **710** to bound the manifold **510** between the flange **515** and the sleeve **520**. As the bolt **350** pushes an ammunition round **270** forward from FIGS. 7A to 7B, the tab **570** pulls along a slide rod **740** from the aft end of the upper half-cylinder **250** to its fore end, thereby providing a cavity within an air cylinder **750** in the air loader **340**.

[0028] The sleeve **520** provides a high-pressure gas charge volume **760**, and the flange **530** provides an axial cylindrical gap **770**. To reset pressurized gas transfer from FIGS. 7B to 7C, the plug **540** axially translates within the volume **760** as the piston **720** attached thereto slides within the cylinder **710**. The ring **550** within the internal volume **760** and mounted to the flange **530** by bolts **545** provides a sealing surface that drives the round **370** along the barrel **190**, while enabling the plug **540** to slide fore and aft to correspondingly close or open the passage between the volume **760** and the gap **770** that reaches the bolt carrier **320**. Compressed gas migrates from the container **240** and the volume **760** within the sleeve **520** into the gap **770** in order to push the bolt **350** forward.

[0029] The ring **550** is screwed to the flange **530** to secure a rubber sealing gasket, thereby sealing the high pressure gas charge volume **760**. The ring **550** forms a gasket under compression. The manifold **510** constitutes a commercial self-contained airtight unit used in industrial applications. Bolting the manifold **510** to the sleeve **520** compresses an o-ring at the interface between the sleeve **520** and the manifold **510**. The o-ring inhibits gas leakage out from the sleeve **520**. The container **240** is a threaded non-pressurized item that enables a stock to thread to the rifle **110, 230**.

[0030] The bolt **350** is not actuated by gas pressure via the ring **550**. Instead, the bolt **350** is actuated using the air loader **340** and tab **570**. The bolt **350** is hollow, enabling gasses to flow therethrough from volume **760** through gap **770** to drive a solid or liquid projectile. The air loader **340** has substantially similar function to manifold **510**, albeit having distinctive purposes and sizes.

[0031] FIGS. 8A and 8B show detail cross-section elevation views of the sUDS rifle assembly **230** in the second configuration for solid rounds **370**. After a round **370** feeds into the chamber **420** from the magazine **220**, the bolt **350** slides forward pushing the round **370** into the barrel **190**. A solenoid **810** operates inside a space **820** within the lower half-cylinder **260**. The space **820** is located within the hand grip **560**. It has a geometrically compensating purpose, as the rifle **110, 230** doesn't quite align with an operator's eye when being shouldered absent the increased height of the handle **560**, which produces the gap **820**.

[0032] FIGS. 9A, 9B and 9C show detail cross-section elevation views of the sUDS rifle assembly **610** in the second configuration for liquid ammunition in the liquid ammunition magazine **630**. The refill port **640** enables addition of liquid ammunition to replace that previously expended from the magazine **630**.

[0033] There is a range of liquid ammunition that can be used, including non-Newtonian fluids, such as corn starch+cotton fibers+water. This mixture appears to maintain shape adequately, even when fired at high speed. The volume **630** is sized to accommodate ammunition. The chamber **420** pushes all the fluid in volume **630** upwards and immediately the bolt **350** drives this slug forward, pushing the fluid into the barrel **190**. Then the air reservoir **245** releases high pressure gas to drive the round **370** out of the barrel **190**. The liquid ammunition requires an additional solenoid **820** that drives the chamber **420** upwards, but the entire firing sequence is the same as for solid projectiles **370**.

[0034] Exemplary embodiments provide a launcher device from which to shoot liquid or solid projectiles at a high speed using gas pressure. Gas pressure encompasses both compressed gas reservoirs and chemically generated initiators. A "liquid projectile" refers to a discrete mass of fluid

from the magazine **620**, unlike a fire hose, which shoots an uninterrupted jet of water. Discrete slugs of fluid are less susceptible to in-flight disintegration than a continuous stream.

[0035] FIGS. **10A**, **10B** and **10C** show detail cross-section elevation views **1000** of the loaded sUDS rifle **230** featuring the firing operation. A gas source **1010** supplies high pressure gas at between 690 kPa to 2.1 MPa. A firing circuit **1020** with a switch **1025** receives electrical power from a power supply **1030**. The switch **1025** is shown as open and closes in response to pushing the trigger **650**. The source **1010** connects to input ports of a pair of solenoid-manifolds **1040** that supplies the air loader **340** and manifold **1050** that refills the container **240** via conduits **1060**.

[0036] The first manifold **1040** connects respective fore and aft conduits **1070** and **1075** to the air loader **340**. The second manifold **1050** includes **1080** and **1085** to the container **240**. The circuit **1020** connects fore and aft conduits to the manifolds **1040** and **1050** via electrical leads **1090**. Upon closure of the switch **1025**, the circuit **1020** energizes the manifolds **1040** and **1050**.

[0037] FIG. **10A** illustrates the rifle **110** in the initial position with the switch **1025** open and a round **370** from the magazine **120** in the chamber **420**. The fore conduit **1070** and aft conduit **1085** contain high pressure gas, while the aft conduit **1075** and fore conduit **1080** are dormant. Note that the source **1010** and power supply **1030** are omitted from the remaining views to reduce clutter.

[0038] FIG. **10B** illustrates the rifle **110** in the load position with the switch **1025** closed in response to the trigger **650**. The first manifold **1040** exchanges fore and aft conduits **1070** and **1075** so the former becomes dormant and the latter fills with high pressure gas. Consequently, the tab **570** pulls the rod **740** and the bolt **350** forward from the loader **340** and the carrier **320**, respectively, thereby pushing the round **370** into the barrel **190**. The circuit **1020** with the switch **1025** open is omitted from the remaining views to reduce clutter.

[0039] FIG. **10C** illustrates the rifle **110** in the firing mode, in which the second manifold **1050** exchanges fore and aft conduits **1080** and **1085** so the former fills with high pressure gas while the latter becomes dormant. Consequently, the plug **540** retracts from the ring **550** to pneumatically connect the volume **760** of the reservoir **245** with the gap **770**. Meantime, the switch **1025** reopens. Following this operation, the rifle **110** returns to reload and recharge mode. This returns the mechanisms to the positions in the initial mode of FIG. **10A**, including the conduits **1070**, **1075**, **1080** and **1085**, with another round **370** pushed into the chamber **420**.

[0040] When the plug **540** retracts, a direct pathway for all the high pressure gasses stored in volume **760** enables immediate discharge, and these gasses flow past the gap **770** and through the bolt **350**. Then, high pressure gas directly acts on the aft side of the round **370**, driving the round **370** out of the barrel **190**.

[0041] The trigger **650** closes an electrical switch **1025**, which signals the solenoid-manifold **1050** to shift high pressure air from fore air conduit **1070**, which restrains the bolt **350**, permitting rounds **370** to ingress to aft air conduit **1075**, which pushes the bolt **350** forward and thereby presses a round **370** forward to be fired. The default position of the bolt **350** is aft. The manifold **1050** converts the trigger pull into switching on which port **1070** or **1075** that high pressure gas is applied, which controls whether the air loader **340** is aft or forward. When the bolt **350** slides forward, the manifold **1050** automatically commands the manifold **510** to pull aft (i.e., backward), which then releases all of the high pressure gas in reservoir **245** all the way through the bolt **350**, and pushes the round **370** out of the barrel **190**.

[0042] The default position for this air cylinder **750** is forward, which prevents high pressure gas from escaping the reservoir **245**. Upon pulling the trigger **650**, first the manifold **1050** tied to the air cylinder **750** fires, with the round **370** pushed into the barrel **190** while the air cylinder **750** remains forward, and second the manifold **1050** tied to the manifold **510** fires, with high pressure gas driving the round **370** out of the barrel **190**, thereby “firing” the rifle **110**, **230**. Then both solenoid **810** and manifold **1050** return to their default positions. The air loader **340** slides backwards and the manifold **510** slides forward, enabling the rifle **110**, **230** to reload.

[0043] Most rifle configurations shoot only one type of ammunition: solid projectiles or slugs of

liquid consistency. A traditional powder driven cartridge, a Nerf blaster, and a tranquilizer dart gun are good examples that span the gamut of devices that launch a solid body using pressurized gas. A toy water gun provides an example of the liquid ejectors. By contrast design of the exemplary rifle enables an operator to shoot both liquid or solid projectiles from a single device.

[0044] Examples of traditional designs include the following. Traditional rifle/sidearm designs are composed of several items: a solid bullet, a cartridge case, a reactive powder, and a primer. The solid bullet, reactive powder, and primer are all located in the cartridge case. Upon mechanical action, a primer is initiated that in turn ignites the reactive powder. High-pressure gases are produced that drive the solid bullet down a barrel at a high speed; no external driving forces are required.

[0045] For example, paintball guns shoot a small solid round using compressed gas and a valve: when the valve opens a small charge of air rapidly expands and drive the paintball round out of the barrel; an external source of compressed gas is required. Water pistols use compressed air to drive a jet of water out of a reservoir using a similar method. However, no weapon design has been found that can shoot both solid and liquid rounds.

[0046] A rifle that can only shoot solid or liquid rounds limits the operator's ability to scale with a threat. For example, for police to engage riot control, they must select between either (a) a large vehicle capable of supporting a continuous jet of water (requiring tremendous infrastructure to be transported) or (b) traditional non-lethal means of crowd control (e.g., bean bags, foam rounds). An individual operator cannot switch between round types.

[0047] In some cases, option (a) will not be operable, and the crowd will overpower the vehicle. On the other hand, traditional non-lethal rounds tend to cause severe injuries. Exemplary embodiments provide law enforcement an opportunity to escalate the intensity of their crowd control weapon as required, and minimizes the likelihood of applying an over- or under-application of force.

[0048] The exemplary sUDS rifle **110**, **210** is an open-bolt design. The firing process proceeds as follows: [0049] (1) The round, whether solid **370** or liquid (such as a slurry from corn starch,

cotton fiber and water) is pushed into the chamber **420** in FIG. **9A**. [0050] (2) The operator depresses the trigger **650**. [0051] (3) Immediately after, an electrical signal initiated by the trigger **650** is sent to a gas solenoid **810**, which pushes the bolt **350** forward. The bolt **350** remains in the forward position. [0052] (4) As a result of step (3), the bolt **350** slides forward pushes the round **370** into barrel in FIG. **7B**. [0053] (5) Immediately after the bolt **350** is in the forward position, a signal *such as from the trigger **650**, a delay circuit or a microcontroller) initiated by the trigger **650** is sent to a gas solenoid **810**, which causes the annular sleeve **520** to retract its sealing surface as ring **550** in FIG. **7C**. [0054] (6) After the sealing surface as ring **550** retracts, high-pressure gases in the reservoir **760** in the sleeve **520** rush into the empty volume, and through the hollow bore of the bolt **350**. [0055] (7) The gas pressure builds in the empty volume in the chamber **420** behind the bolt **350**, and begins to drive the round **370** down the barrel **190**. [0056] (8) The round **270** leaves the barrel **190**, as propelled by the compressed gasses from container **240**. [0057] (9) The solenoid valve **810** receives an electrical signal to close the sealing surface as ring **550**. The reservoir **245** connects to a high-pressure gas container **240**, and upon its closure, the reservoir **245** refills with high-pressure gas. from the container **240**. The solenoid valve **810** receives an electrical signal to retract the bolt **350**. [0058] (11) As the bolt **350** retracts, the next round **270** is pushed into the chamber **420**. [0059] (12) The rifle **110** is ready to fire again.

[0060] The most simple design features the trigger **650** and solenoid **810** as an electric circuit, which is open by default until the trigger **650** is depressed as switch **1025**. Meanwhile, the signal also activates a delay circuit, which controls the more complicated designs that engage the trigger **650** and an electrical signal into a microcontroller for more precise timing.

[0061] The method of loading ammunition depends on what is fired: [0062] (1) For solid rounds **370** in a removable magazine **120**, the force exerted by a magazine spring (not shown) pushes a solid round **370** into the chamber **420**. [0063] (2) Liquid rounds do not exist in a coherent form

outside the chamber **420**. The fluid exists in the magazine **630**, and a metered quantity is pumped or pushed into the chamber **420**. Several methods exist to accomplish this. After loading, both round configurations operate exactly the same in the rifle **110, 230**.

[0064] Pneumatic rifles (such as air guns) are not new. Toy guns that shoot non-lethal rounds (Nerf guns) are not new. However, the sUDS rifle **110, 230** is designed to shoot a variety of solid and liquid rounds at velocities at or exceeding 100 m/s, with projectiles that weigh up to and exceeding 20 grams.

[0065] In many ways, one can conceive of the sUDS rifle **110, 230** as a novel mechanism for a single individual to shoot large projectiles—having a variety of purposes—at a high velocity and large distance accurately. There are several variations on the rifle design in both the loading mechanisms, the projectile that is ejected, and the aesthetics of the design.

[0066] A rifle that can shoot both solid rounds and liquid rounds finds many applications. The most pressing issues are addressed herein. Counter small-UAS (c-sUAS) weapons are of considerable interest. Small UASs plague the battlespace, and no non-lethal kinetic or non-Electronic Warfare technique is available for dealing with the problem. For example, soldiers cannot shoot at a quadcopter in a city due to the collateral damage caused by the rifle rounds. Fire hoses are not very effective at long range. Exemplary embodiments solve the c-sUAS problem by permitting a single operator to engage these small UASs with great efficacy without collateral damage to material or personnel.

[0067] In the same vein, there is a huge issue right now with non-lethal weapons for crowd control. Given the contemporary political climate, police need a non-Lethal weapon system that is actually non-lethal. The proposed method can fulfill that need by shooting non-lethal rounds.

[0068] While certain features of the embodiments of the invention have been illustrated as described herein, many modifications, substitutions, changes and equivalents will now occur to those skilled in the art. It is, therefore, to be understood that the appended claims are intended to cover all such modifications and changes as fall within the true spirit of the embodiments.

Claims

1. A hand-held long-gun for interchangeably propel a round of one of solid and liquid consistency, said gun comprising: a gas actuator for storing and releasing compressed gas; a firing module that includes a chamber for receiving said gas to push forward a bolt in response to actuation by a trigger, said module disposed forward of said actuator; a magazine for containing ammunition that constitutes the round; a loader for laterally receiving the round from said magazine and longitudinally receiving said bolt to eject the round, said loader disposed forward of said module; and a barrel for directing the round ejected from said loader, said barrel disposed forward of said loader.
 2. The gun according to claim 1, wherein said gas actuator comprises a compressed air storage container and an air release reservoir.
 3. The gun according to claim 1, wherein said control module comprises an upper portion and a lower portion.
 4. The gun according to claim 1, wherein said magazine can one of attach and detach from said loader.
 5. The gun according to claim 1, wherein said firing module further includes a solenoid to activate in response to said trigger to release said gas from said actuator for forcing said bolt forward.
 6. The gun according to claim 1, further including a handle for housing said trigger and a butt stock for supporting said actuator.
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